

CHEKITHA SWAYAMPU

DATA ANALYST

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SUMMARY

- **Data Analyst** with **4+ years** of **experience** leveraging advanced **analytics, statistical modeling, and data engineering** to improve financial risk accuracy, healthcare outcomes, operational visibility, and enterprise decision-making.
- Skilled in transforming multi-domain datasets using **SQL, Python, and BI tools** to enhance forecasting precision, automate reporting, optimize operational planning, and deliver scalable, data-driven strategic insights.
- Experienced in developing **predictive models, automated KPI dashboards, and end-to-end analytical workflows** to improve revenue quality, strengthen compliance, enhance patient outcomes, and modernize enterprise data ecosystems.

TECHNICAL SKILLS

Languages:	Python, Scala, R, SQL, SAS
Packages/Libraries:	NumPy, Pandas, Matplotlib, Seaborn, ggplot2, SciPy, Scikit-learn, TensorFlow
Visualization Tools:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP)
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook
Database Management:	MySQL, PostgreSQL, SQL, Oracle, Snowflake, dbt
Cloud Technologies:	AWS, Azure
Methodologies:	SDLC, Agile, Waterfall, Predictive Modeling
Version Control:	Git, GitHub, Bitbucket
Data Manipulation:	Data Analysis, Data Mining, Data Preprocessing, Data Mapping, Data Cleaning, Data Visualization, Data Modeling, Data Warehousing, Data Storytelling, Data Wrangling, Data Acquisition, Data Integration, Data Transformation
Other Skills:	SSIS, SSRS, Machine Learning Algorithms, CI/CD, Probability Distributions, Confidence Intervals, ANOVA, Association Rules, Clustering, Advanced Analytics, ETL Processes, Informatica, Report Generation, Regression, A/B Testing, Forecasting & Modeling, Hypothesis Testing, Time Series Analysis, Root Cause Analysis

WORK EXPERIENCE

Data Analyst HCA Healthcare , VA, USA	Aug 2024 – Present
• Spearheaded AI-enabled clinical documentation transformation using ambient intelligence to streamline workflows, enhance EHR accuracy, and deliver actionable insights for patient care, adherence, and population health. • Automated ETL pipelines using Python and Kafka, integrating 15+ pharmacy, claims, and EHR sources, processing 1.6M+ records weekly, reducing manual effort 45%, and producing regulatory-compliant datasets for analytics and decision-making. • Validated Epic EHR data with clinicians, resolving 25% anomalies, improving predictive model accuracy, and generating actionable insights to optimize patient adherence, reduce costs, and enhance population health outcomes. • Optimized PostgreSQL queries on longitudinal patient datasets with indexing and partitioning, reducing latency by 60%, enabling scalable clinical and financial analytics, and supporting real-time reporting for care teams efficiently. • Conducted exploratory data analysis using Python, Pandas, and Seaborn, identifying patient behavior patterns, adherence trends, and drug utilization anomalies to inform predictive models and evidence-based clinical interventions. • Developed predictive models with Scikit-learn, implementing classification and regression algorithms to forecast medication adherence and patient risk, achieving 65% accuracy and supporting proactive population health strategies effectively. • Built 12 Power BI dashboards with DAX and Power Query, visualizing patient outcomes, adherence KPIs, and pharmacy metrics, reducing reporting time by 40% and enabling data-driven operational decisions. • Executed hypothesis testing on adherence interventions across 50,000+ patient records, improving adherence by 20%, informing enterprise-wide care protocols, and enabling measurable improvements in clinical and operational outcomes. • Architected HIPAA-compliant AWS S3 cloud storage, ensuring secure, encrypted, scalable storage for terabyte-scale datasets to support analytics, predictive modeling, and regulatory compliance requirements effectively. • Collaborated with clinicians, pharmacy leads, and data engineers to align analytics with HIPAA, value-based care objectives, and organizational goals, driving measurable patient outcomes and operational efficiency improvements.	

- Delivered analytics enhancements for core banking modernization, improving data reliability, streamlining workflows, and increasing enterprise visibility across deposits, lending, payments, treasury, regulatory, and compliance processes.
- Directed **Python** and **Alteryx** ETL workflows processing **1.2M+** weekly financial records, improving data quality by **34%** and accelerating regulatory portfolio risk assessments across critical banking, audit, and compliance operations.
- Engineered **Scikit-learn** predictive risk models on **120K+** trades, applying regression and ensemble algorithms, achieving **62%** accuracy, enabling proactive high-risk exposure detection and improved capital planning for investment decisions.
- Refined **SQL** pipelines across multi-year account, transaction, and market datasets, reducing query latency by **41%** and supporting analytics review of **300K+** monthly transactions for compliance, audit tracking, and operational reporting.
- Produced **10+ Tableau** dashboards, including leadership views, delivering exposure insights, KPI tracking, liquidity monitoring, and performance metrics using daily refreshed datasets for faster treasury decision-making and risk evaluation.
- Constructed **Azure Data Factory** pipelines with parameterized ingestion for **2TB+** datasets, reducing manual effort, increasing scalability, and enabling enterprise-grade integration across downstream financial reporting and analytics systems.
- Upgraded **Snowflake** warehouse performance using **Informatica ETL** with schema tuning, optimized partitioning, and query improvements, reducing execution times **37%** while enhancing responsiveness for complex portfolio analytics workloads.
- Performed **Advanced Excel** modeling across **80+** portfolios using macros, pivoting, and simulations, improving liquidity forecasting accuracy and enabling stronger capital planning, budgeting alignment, and financial planning decisions.
- Generated standardized **Confluence** documentation detailing pipelines, models, data rules, and dashboards, improving onboarding efficiency, reproducibility, collaboration across enterprise technology, compliance, and risk management teams.
- Unified portfolio, risk, and IT teams to align analytics outputs with **SOX** and **Basel III** standards, strengthening data governance, reducing operational risk, and enhancing regulatory compliance effectiveness.

PROJECTS

1. Temperature Forecasting using Time Series Techniques

- Led a team of 3 to analyze **2.9M** U.S. daily temperature records and forecast short- and long-term trends using machine learning, deep learning, and time series models; applied feature engineering, statistical analysis, and encoding.
- Architected and benchmarked **ARIMA, SARIMA, Random Forest, XGBoost, and LSTM** forecasting models using **RMSE, MAE, and R²**; identified **XGBoost (RMSE 3.74 °F, MAE 2.87 °F)** as most accurate, reducing error by **44%**.
- Visualized forecast results and actual vs predicted trends using **Power BI** and **Plotly** for easy stakeholder interpretation.

2. Impact of Weather on Energy Consumption Using AWS Cloud Services

- Implemented a predictive analysis model in **AWS SageMaker**, leveraging historical weather datasets to forecast regional energy consumption patterns, integrating machine learning workflows.
- Built a data pipeline using **AWS Glue** for ETL and **S3** for storage, enabling efficient data ingestion and preprocessing; visualized energy consumption trends and model predictions using **AWS QuickSight**, providing actionable insights.

3. Starbucks Customer Segmentation and Offer Success Prediction

- Performed customer segmentation on **17,000+** Starbucks application users with **K-Means clustering** using demographics and offer interaction data, revealing patterns that informed personalized campaign strategies.
- Developed and fine-tuned predictive models to estimate customer response to BOGO, Discount, and Informational offers; achieved up to **97% accuracy**, enabling more effective targeted marketing.

CERTIFICATIONS

- **AWS Certified Data Engineer - Associate**
- **Oracle Analytics Cloud 2025 Certified Professional**
- **Oracle Cloud Infrastructure 2025 Certified Data Science Professional.**

EDUCATION

Master of Science in Data Science

- George Washington University, Washington, D.C., USA

Bachelor of Technology in Computer Science Engineering with specialization in Data Analytics

- Vellore Institute of Technology, AP, India